

LeWAM: END-TO-END WORLD AND ACTION MODELING WITH JEPAS

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ABSTRACT

Joint-embedding predictive architectures (JEPAs) learn compact latent world models without pixel reconstruction. Existing JEPA pipelines typically train visual representations and action-conditioned dynamics before adding a control policy or solver. We introduce **LeWAM**, an end-to-end JEPA for control with: (1) a joint training objective over next latent state prediction and action flow matching, and (2) an isotropic Gaussian constraint on the latent state, enabling a single model to support direct policy execution and efficient planning by evaluating sampled actions with its learned latent dynamics. Compared with prior JEPAs in visual planning, LeWAM improves success rate from 28.6% to 89.7% on contact-rich tasks and from 10.9% to 46.2% on long-horizon tasks, and plans $32.3\times$ faster. On behavior cloning tasks in simulation and physical robots, LeWAM is also competitive with established methods. Interpretability analyses show that LeWAM learns control-relevant representations and action distributions effectively guiding test-time planning and control.

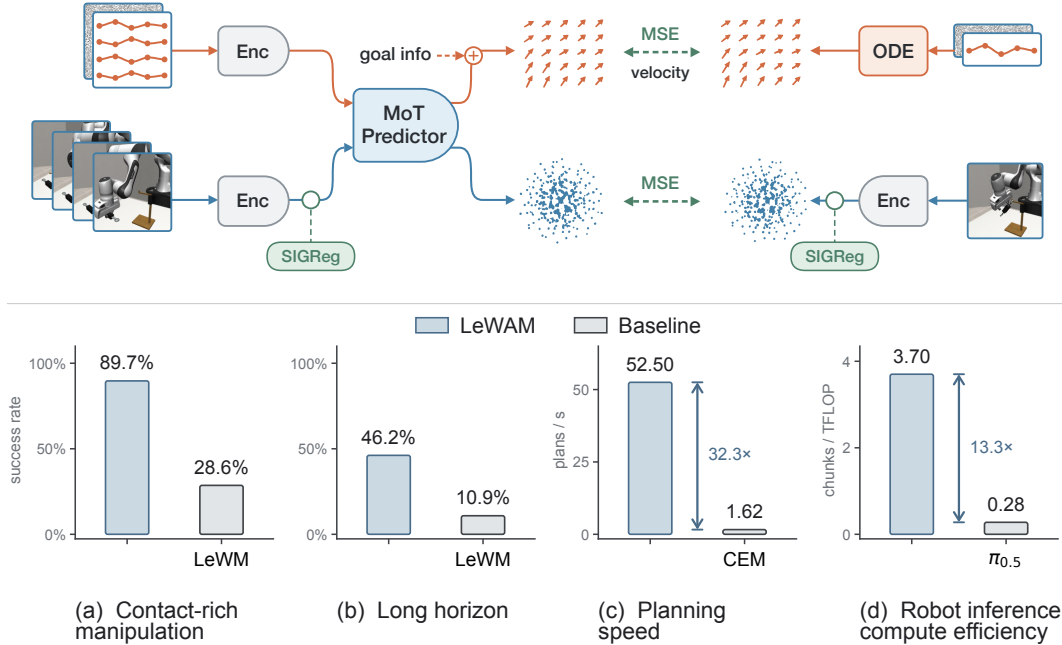


Figure 1: LeWAM Training. A shared encoder encodes observed frames with SIGReg (Balestrierio & LeCun, 2025) regularization. The Mixture-of-Transformers (Liang et al., 2025) learns latent-state prediction from clean actions and action flow matching from noisy actions with goal information. **Results Overview.** The mean of (a) planning success on contact-rich tasks; (b) planning success on a 100-step goal offset; (c) plans/s versus CEM (Rubinstein, 1999); (d) action chunks per TFLOP versus $\pi_{0.5}$ (Physical Intelligence et al., 2025). Full results are in Section 4.

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LeWM, V-JEPA 2	SIGReg, DINOv2	DreamZero, Fast-WAM	CEM, MPPI, iCEM	Evaluation Tasks
World-model training post-trained control	Latent regularization pretrained features	Generative Video WAMs	Population-based action search	Goal-reaching evaluation
LeWAM: End-to-End World and Action Modeling with JEPAs				
✓ Unified world model and control	✓ Control-relevant representation	✓ No reconstruction ✓ No pre-training	✓ Policy-guided search ✓ Gradient-based refinement	✓ Contact-rich ✓ Long horizon

Figure 2: **Positioning of LeWAM.** The columns place LeWAM alongside related approaches to world-model training, visual representation learning, world action modeling, and action search. LeWAM jointly trains latent dynamics and a policy through a shared encoder, predicts in latent space without reconstruction, and refines learned action proposals through its dynamics. The final column summarizes its evaluation on contact-rich tasks and longer goal horizons.

1 INTRODUCTION

Visual control requires autonomous agents to complete tasks by selecting actions based on visual observations of the world. World models address this challenge by predicting the consequences of candidate actions, allowing agents to anticipate how the world will evolve before executing them. Joint-embedding predictive architectures (JEPAs) (LeCun, 2022) offer a compact world modeling paradigm where future world states are predicted in latent space rather than by reconstructing pixels. Recent works such as DINO-WM (Zhou et al., 2024) and LeWorldModel (LeWM) (Maes et al., 2026b) demonstrate the promise of applying action-conditioned latent world models to downstream control. These pipelines obtain visual features from pretrained encoders (Zhou et al., 2024; Assran et al., 2025), or train representations from scratch with distributional regularizers such as SIGReg (Balestriero & LeCun, 2025; Maes et al., 2026b) to prevent collapse. However, these methods alone do not reliably retain control-relevant information; JEPAs can favor temporally persistent distractors (Sobal et al., 2022), while action-conditioned dynamics may retain extraneous observation information (Brandfonbrener et al., 2023).

Control with these models typically requires an online solver such as the cross-entropy method (CEM) (Rubinstein, 1999) or a separately trained action predictor. When initialized with a broad Gaussian prior over action sequences, CEM demands extensive rollouts to identify viable candidates, particularly as action dimensionality or planning horizon grows (Huang et al., 2021). Contact transitions also complicate local action optimization because the dynamics can switch between modes (Pang et al., 2023). A post-trained action predictor can provide direct control, as in GC-IDM (Nguyen et al., 2026), or guide action search, as in PRISM (Wang et al., 2026). Training these predictors on frozen JEPA embeddings leaves action supervision disconnected from visual representation learning.

We introduce **LeWAM**, an end-to-end JEPA that jointly learns representation, action-conditioned dynamics, and a stochastic policy in a single training stage (Figure 1). Training combines two ingredients: a joint training objective over next-latent-state prediction and action flow matching, and SIGReg (Balestriero & LeCun, 2025) to regularize the latent states. A Mixture-of-Transformers (MoT) (Liang et al., 2025) predictor uses clean actions for latent dynamics prediction and noisy actions for policy learning through flow matching, so both objectives train the shared visual representation. Consequently, LeWAM supports both direct policy execution and visual planning by refining action sequences sampled from its learned policy. Our contributions are threefold:

- To our knowledge, the first JEPA that jointly learns representation, action-conditioned latent dynamics, and stochastic policy, using a joint prediction objective, one weight for latent-space regularization, and approximately 17M parameters.
- A planning procedure that jointly imagines the consequences of actions and how to act in the resulting states, which samples candidate action sequences from the learned action distribution and refines them with gradient descent through learned dynamics.
- Across ten heterogeneous simulated environments, LeWAM improves goal-reaching success over LeWM (Maes et al., 2026b), particularly on contact-rich tasks. Four-task comparisons extend

these gains to longer goal horizons and show faster planning. On goal-agnostic task completion in simulation and physical robot execution, LeWAM performs competitively with established baselines. Linear probes reveal decodable physical-state and action information, recorded-action rollouts visualize its learned dynamics, and planner ablations measure the gains from its stochastic action proposals.

2 RELATED WORK

World Models. World models (Ha & Schmidhuber, 2018) learn environment dynamics to enable agents to simulate candidate action outcomes for planning and control. Early approaches like PlaNet (Hafner et al., 2019) perform planning via latent rollouts, whereas TD-MPC (Hansen et al., 2022) and TD-MPC2 (Hansen et al., 2024) ground latent dynamics in reward and value predictions. Dreamer (Hafner et al., 2020; 2023) learns generative latent dynamics with pixel reconstruction, while VideoGPT (Yan et al., 2021) and IRIS (Micheli et al., 2023) model discrete visual tokens. LeWAM learns latent dynamics and a policy directly from offline trajectories, without reward supervision or pixel reconstruction.

Joint-Embedding Predictive Architectures. JEPAs predict target states in representation space without pixel reconstruction, using variance and covariance constraints (Bardes et al., 2022), momentum targets (Assran et al., 2023; Bardes et al., 2024), or distributional regularization (Balestrierio & LeCun, 2025; Wu et al., 2026) to prevent collapse. For action-conditioned world modeling, DINO-WM (Zhou et al., 2024) and V-JEPA 2-AC (Assran et al., 2025) build on pretrained representations, whereas LeWM (Maes et al., 2026b) leverages SIGReg (Balestrierio & LeCun, 2025) to learn the encoder and latent dynamics from scratch. Additional action prediction objectives attempt to make representations action-aware in PLDM (Sobal et al., 2025) or as an anti-collapse signal in SMWM (Ivashkov et al., 2026). LeWAM jointly trains action-conditioned latent dynamics and policy in one predictor, so both consequence prediction and action supervision shape the shared representation.

Control with Latent Dynamics. Conventional planning in latent world models uses population-based solvers, such as CEM (Rubinstein, 1999), iCEM (Pinneri et al., 2021), MPPI (Williams et al., 2017), or gradient-based methods (Henauff et al., 2017; Huang et al., 2021). Population-based search can require many model rollouts as the action dimension or horizon grows (Pinneri et al., 2021; Huang et al., 2021), and contact mode changes in manipulations also complicate local optimization (Pang et al., 2023) (more discussion in Appendix E). GC-IDM (Nguyen et al., 2026) post-trains a goal-conditioned action predictor for direct control, whereas PRISM (Wang et al., 2026) trains a Gaussian prior head on frozen world-model features; its predicted mean and standard deviation guide MPPI sampling. LeWAM jointly trains the latent world model and a flow-based action predictor, using its sampled actions for direct control or gradient-based planning.

World Action Models. These models couple world prediction with action prediction (Table 13, Appendix E). Built on pretrained video generative models, DreamZero (Ye et al., 2026) jointly models future video and actions, while Fast-WAM (Yuan et al., 2026) retains video co-training but skips future video generation at inference. UVA (Li et al., 2025), UWM (Zhu et al., 2025), and Cosmos 3 (NVIDIA, 2026) specifically support both forward dynamics modeling (FDM) and policy learning, but optimize only one objective per sample, rather than both jointly on the same pass. LeWAM jointly trains its encoder and predictor end-to-end using latent-state prediction and action flow matching, without pixel reconstruction. Each training sample supplies clean actions for FDM and noisy actions for stochastic policy modeling. LeWAM learns action-conditioned dynamics from expert and non-expert interactions and uses them to evaluate and refine actions at test time.

3 METHODOLOGY

Given reward-free offline trajectories of pixel observations $\mathbf{o}_{1:T}$ and actions $\mathbf{a}_{1:T}$, we jointly train a visual encoder and a predictor including action-conditioned dynamics and a flow matching policy, supporting direct control and planning with action-conditioned latent dynamics.

3.1 END-TO-END TRAINING WITH LEWAM

Architecture. Figure 1 summarizes the model at a high level. Let $\mathbf{z}_t = \text{Enc}(\mathbf{o}_t)$ denote the latent representation of observations $\mathbf{o}_t$. All time indices refer to raw environment steps, such that action $\mathbf{a}_t$ transitions $\mathbf{o}_t$ to $\mathbf{o}_{t+1}$. We use a frameskip of K environment steps, so future observations used as dynamics targets are spaced at K -step intervals. In contrast, the policy models a contiguous action chunk $\mathbf{A}_{t,C} = (\mathbf{a}_t, \dots, \mathbf{a}_{t+C-1})$ with $C \geq K$. For visual planning, we assume access to goal information $\mathbf{c}_t$ that contains the goal image embedding $\mathbf{z}_g$ and its normalized remaining horizon from $\mathbf{z}_t$.

The encoder consists of a ResNet-18 backbone (He et al., 2016), spatial softmax pooling, and a multilayer perceptron (MLP) with two projection layers. The same encoder processes all the pixel observations: history, future target, and goal image. The predictor is comprised of a state branch f_z and an action branch f_a . Specifically,

$$\begin{aligned}\hat{\mathbf{z}}_{t+K} &= f_z(\mathbf{q}_{t+K}^z; \mathbf{z}_{\leq t}, \mathbf{A}_{t,K}), \\ \hat{\mathbf{v}}_{t,C} &= f_a(\mathbf{A}_{t,C}^{(s)}; s, \mathbf{z}_{\leq t}, \mathbf{c}_t)\end{aligned}\quad (1)$$

where $\mathbf{A}_{t,C}^{(s)}$ is a noisy C -action chunk at flow timestep $s \in [0, 1]$ (see Equation 2) and $\mathbf{q}_{t+K}^z$ is a learned query token specifying the future state to predict. Integrating the predicted velocity field $\hat{\mathbf{v}}_{t,C}$ yields the predicted action chunk $\hat{\mathbf{A}}_{t,C}$.

To simultaneously balance latent-prediction and flow-matching objectives while allowing f_z and f_a to share information about states and actions, we use the Mixture-of-Transformers (MoT) (Liang et al., 2025) architecture. That is, each branch applies independent query, key, value, and feed-forward projections for its respective modality (latent states or actions) while sharing one masked self-attention. Figure 3 specifies the information available to each branch during attention. The state query $\mathbf{q}_{t+K}^z$ reads history latents and clean actions, and noisy action tokens $\mathbf{A}_{t,C}^{(s)}$ read history latents and their own causal prefix. History latents attend only to themselves, blocking indirect clean-action leakage to f_a . Goal information conditions the noisy action tokens; the dynamics branch reads history latents and clean actions. Appendix A gives the full token-level mask, layer sizes, and training settings.

Training Objective. For flow matching (Lipman et al., 2023), given a recorded action chunk $\mathbf{A}_{t,C}$, sample s uniformly from $[0, 1]$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then construct

$$\mathbf{A}_{t,C}^{(s)} = (1-s)\epsilon + s\mathbf{A}_{t,C}, \quad \mathbf{v}_{t,C} = \frac{d\mathbf{A}_{t,C}^{(s)}}{ds} = \mathbf{A}_{t,C} - \epsilon \quad (2)$$

The action branch regresses the path velocity $\mathbf{A}_{t,C} - \epsilon$ from $\mathbf{A}_{t,C}^{(s)}$. The latent dynamics and action flow matching losses use mean squared error (MSE), averaged over the scalar coordinates of each prediction:

$$\begin{aligned}\mathcal{L}_{\text{dyn}} &= \mathbb{E}_{\mathcal{D}}[\text{MSE}(\hat{\mathbf{z}}_{t+K}, \mathbf{z}_{t+K})], \\ \mathcal{L}_{\text{act}} &= \mathbb{E}_{\mathcal{D}, s, \epsilon}[\text{MSE}(\hat{\mathbf{v}}_{t,C}, \mathbf{A}_{t,C} - \epsilon)].\end{aligned}\quad (3)$$

We use SIGReg (Balestriero & LeCun, 2025) to preserve latent diversity through an isotropic Gaussian distributional constraint. Let $\mathbf{Z}$ collect the encoded history and target observations in a mini-batch. With M random unit directions $\mathbf{u}^{(m)}$ and the univariate Epps–Pulley (Epps & Pulley, 1983) test statistic T against $\mathcal{N}(0, 1)$, the regularizer is

$$\mathcal{L}_{\text{sig}} = \frac{1}{M} \sum_{m=1}^M T(\mathbf{Z}\mathbf{u}^{(m)}). \quad (4)$$

With regularization weight λ_{sig} , the full objective is

$$\mathcal{L}_{\text{LeWAM}} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{act}} + \lambda_{\text{sig}}\mathcal{L}_{\text{sig}}. \quad (5)$$

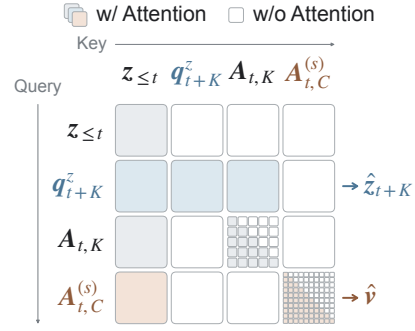


Figure 3: **MoT Attention.** Filled cells permit attention, while white cells are masked. The state query reads history and clean actions, and noisy action tokens read state history and their own causal prefix. Small grids show causal attention. Arrows mark the state branch and action velocity branch.

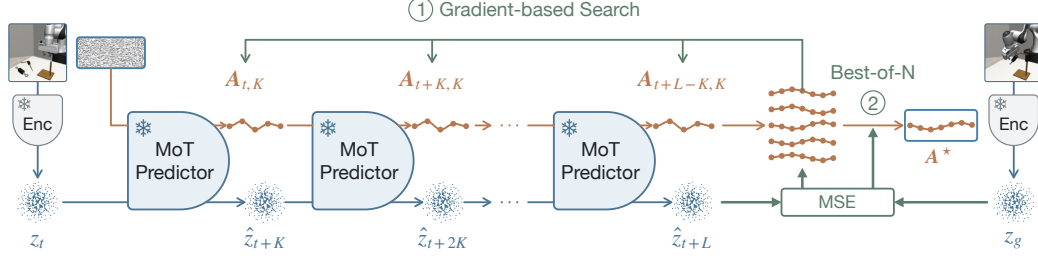


Figure 4: **Planning with LeWAM.** Each MoT predictor stage samples N action chunks from Gaussian noise and uses the first K actions in each chunk, denoted $A_{t,K}$, to advance the latent dynamics from step t to $t + K$. Five-point curves depict action chunks (we use $K = 5$), and stacked curves denote N sampled plans of length L . Step 1 refines all plans by gradient descent through the frozen LeWAM dynamics using the goal-state cost. Step 2 selects the lowest-cost refined iterate as the final plan, shown as A^* . Appendix A.1 gives the full procedure in a pseudo-algorithm.

3.2 EFFICIENT CONTROL WITH LEWAM

LeWAM supports both direct control through its learned policy and planning through rollouts that alternate action sampling and latent-state prediction.

Direct Control. With history and optional goal context fixed, one C -action chunk is sampled by integrating along the learned velocity field over the interval $[0, 1]$ from Gaussian noise:

$$\frac{dA_{t,C}^{(s)}}{ds} = f_a(A_{t,C}^{(s)}; s, z_{\leq t}, c_t), \quad A_{t,C}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \hat{A}_{t,C} = A_{t,C}^{(1)} \quad (6)$$

The first K actions are executed and resampling begins from the resulting observation.

Planning. LeWAM uses a gradient-based refinement strategy over a population of action candidates sampled from the learned action distribution (Figure 4). To simplify notation, let $A = (a_t, \dots, a_{t+L-1})$ denote a plan to reach the goal z_g after L raw environment steps from time t , where L is a multiple of frameskip K .

Starting from the observed history $z_{\leq t}$, one population of N candidate plans is generated from the learned action distribution by alternating action sampling and latent dynamics. Specifically, at each step, Eq. (6) generates N action chunks from the observation history and goal. The forward dynamics branch uses the first K actions of each chunk to predict the N corresponding next states. Each predicted state is appended to the fixed-length history of its candidate, the planning horizon is updated, and the process continues for L total steps. This leaves N action sequence candidates $\{A^{(n)}\}_{n=1}^N$. Starting from $\{A^{(n)}\}_{n=1}^N$, we refine each candidate by gradient descent through latent dynamics (Henaff et al., 2017; Jyothish S V et al., 2023) with respect to the latent cost-to-goal

$$\mathcal{C}(A) = \text{MSE}(\hat{z}_{t+L}(A), z_g) \quad (7)$$

With learning rate η , the goal-cost update takes the gradient-descent form

$$a_{t+k} \leftarrow a_{t+k} - \eta \frac{\partial \mathcal{C}}{\partial a_{t+k}}, \quad k = 0, \dots, L-1, \quad (8)$$

The descent iterate with the lowest $\mathcal{C}$ across all candidates is chosen. In practice, we add a small proximity penalty to discourage large changes from the sampled plans (Appendix A.1). We execute the first K actions, observe the resulting state, and repeat sampling and refinement until the goal is reached, as in receding-horizon model predictive control.

4 EXPERIMENTS

Our experiments aim to address the following questions: (1) Can a single LeWAM architecture learn stably from offline data and support visual planning across heterogeneous tasks and settings? (2) Does LeWAM jointly learn informative action-conditioned latent dynamics, control-relevant representations, and multi-modal distributions over actions? (3) Can LeWAM further support behavior cloning in simulated environments and physical-robot control?

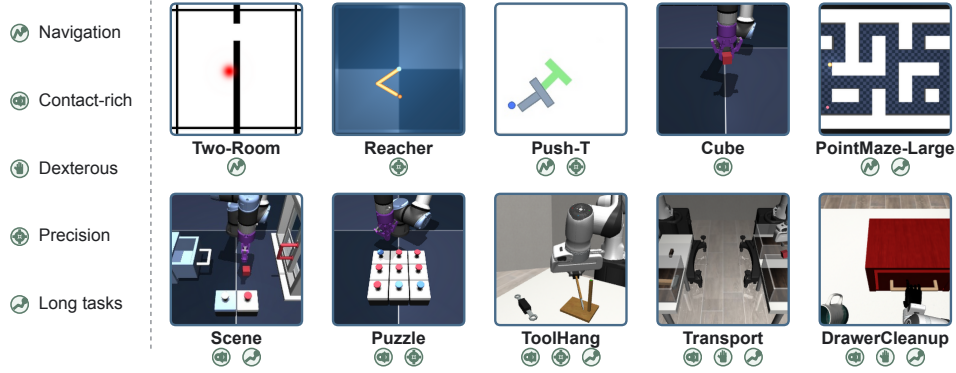


Figure 5: **Simulation Environments.** Symbols beneath each frame denote overlapping task attributes, with their meanings shown in the legend to the left.

4.1 EVALUATION PROTOCOL

Heterogeneous Simulated Environments We evaluate control across ten simulated navigation and manipulation environments (Figure 5). The environments vary in task functionality and temporal horizon, and their datasets include expert demonstrations, random-policy interactions, and mixed-quality data. Four environments adapt the benchmarks used in LeWM (Maes et al., 2026b). The other six add challenging contact-rich manipulation and navigation. Appendix B details task construction, data, and success criteria. For *Cube*, success checks both cube and end-effector positions.

Evaluation Protocols. We distinguish two separate evaluation settings. Visual planning assumes access to a goal image, and a goal-conditioned policy or planner must reach the depicted state from its current (random) starting position within a fixed step budget. Task completion provides no goal image and evaluates behavior cloning from demonstrations: policies execute actions from start to finish to complete one specific task. Within each setting, methods share training and evaluation data, observations, action limits, goal offsets, execution budgets, and success predicates. Appendix B.2 details goal construction, evaluation conditions, and success criteria.

4.2 VISUAL PLANNING WITH LEWAM

We compare with four goal-conditioned baselines spanning latent-dynamics planning, an inverse-dynamics policy on a frozen world model, and direct policy execution:

- **LeWM (CEM)** (Maes et al., 2026b). Learning compact action-conditioned latent dynamics end-to-end with SIGReg, planning with CEM over learned latent dynamics.
- **LeWM (IDM)** (Nguyen et al., 2026). Post-trains a goal-conditioned IDM on frozen LeWM.
- **DINO-WM** (Zhou et al., 2024). Learns dynamics on a frozen DINOv2 encoder with CEM planner.
- **GCBC** (Ghosh et al., 2021). Trains a goal-conditioned policy on frozen DINOv2 features.

Figure 6 reports goal-reaching success across ten environments. Across the ten tasks, LeWAM averages 93.5% success versus 50.9% for LeWM (CEM) (Maes et al., 2026b), a 42.7 percentage-point gap. LeWAM exceeds that baseline on every task, with particularly large gaps on contact-rich ToolHang (94.7 versus 35.4) and Transport (83.3 versus 2.8). On the logically structured Puzzle task, LeWAM reaches 80.7% success versus 44.0% for goal-conditioned GCBC (Ghosh et al., 2021).

Long-Horizon Planning. Using the same checkpoints, we move the goal farther along each recorded trajectory while keeping its starting frame fixed on Push-T, Cube, Scene, and Puzzle. Unlike navigation, these tasks introduce significantly higher behavioral diversity as the planning horizon grows. At a 100-step goal offset, LeWAM achieves 46.2% mean success across these four tasks versus 10.9% for LeWM (Figure 7).

Towards Scalable Planning. Beyond accuracy gains, LeWAM delivers planning acceleration through two core mechanisms: (1) an expressive distribution prior that narrows the search space, and (2) gradient-based optimization that leverages GPU’s parallel architectures to unlock scalable test-time search. Across four visual-goal tasks, it averages 52.50 plans/s versus 1.62 plans/s for LeWM (CEM), a 32.3 $\times$ speedup (Table 6; Figure 1(c)).

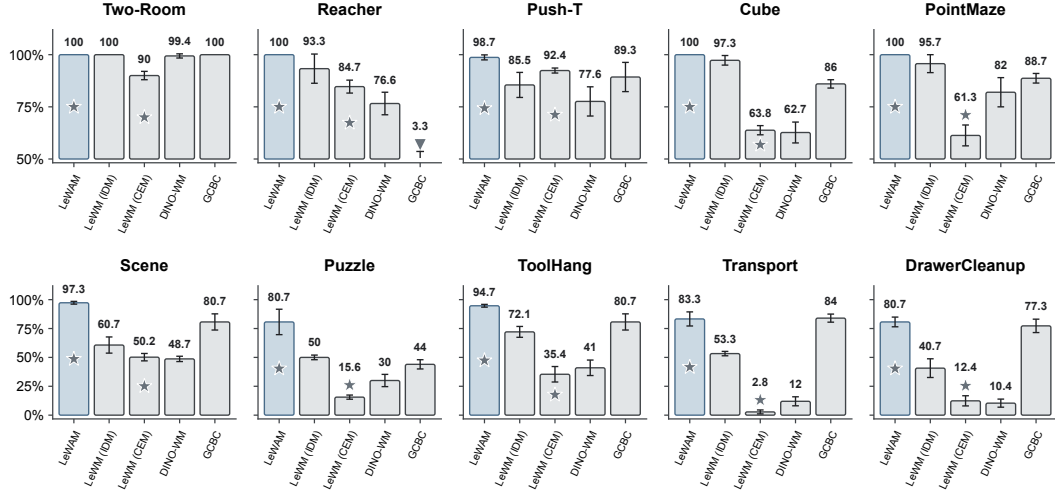


Figure 6: **Visual Planning across Ten Simulated Environments.** Most panels report the success rate under a goal offset of $L = 25$ environment steps and permit $2L = 50$ actions across 3 seeds. For **Two-Room**, we use 100/150 action steps following LeWM setting; For LeWM (CEM), we report 10 seeds for stability. Details are present in Appendix B.2. Error bars denote one standard deviation; ★ indicates models with end-to-end training. Linear axes start at 50% for the first five panels.

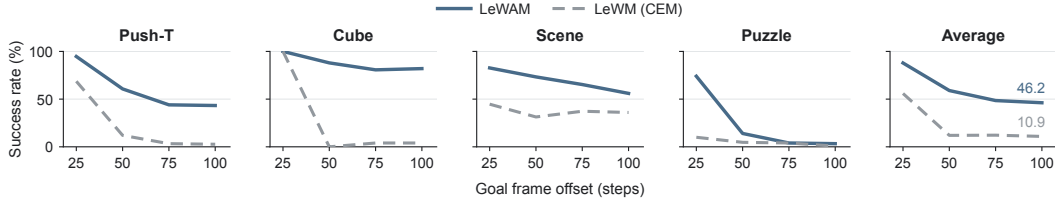


Figure 7: **Visual Planning Success across Different Horizons.** The task panels show mean success rates across 3 seeds. Each sweep starts at the first trajectory frame but with different goal offsets.

4.3 UNDERSTANDING LEWAM’S DYNAMICS, REPRESENTATIONS, AND ACTIONS

Latent Representation Probing. To better measure what information is learned by our joint-training objective, we evaluate how well linear probes recover physical states, proprioceptive information, and actions from LeWAM’s representations. As highlighted in Table 1 and Appendix C.3, training with an action modeling objective noticeably improves LeWAM’s linear action and proprioception decodability compared to LeWM, increasing average proprioception, action, and IDM R^2 by +0.21, +0.20, +0.19 respectively (see Table 9, Appendix C.3).

The Role of SIGReg. SIGReg (Balestriero & LeCun, 2025; Maes et al., 2026b) constrains the latent distribution to prevent representation collapse. Without it, latent-only LeWM yields $R^2 = 0.00$ on every reported **Push-T** probe, while LeWAM with either action head retains decodable state and action information (Table 1). This supports action supervision as an *anti-collapse signal*: observed actions anchor the learned representation to targets independent of the encoder. However, only constraints from action prediction is not enough, beyond anti-collapse, we find that SIGReg can largely improve planning: on **Push-T**, the planning gain is 14.7 percentage points with SIGReg, compared to 2.7 without it (Table 7). The probes in Tables 1 and 9 show that SIGReg improves object-state decodability as well. This is because SIGReg can retain informative latent representation and dynamics which help downstream planning, and make latent distance-to-goal more sensitive to changes in object state.

Figure 8(a) compares full LeWAM with its no-SIGReg ablation. Removing SIGReg lowers planning success in the three tasks shown, most strongly on **Push-T** (25.3 percentage points), followed by **Puzzle** (5.3) and **Scene** (4.0).

Table 1: Linear probing and the role of SIGReg. All linear probes report R^2 score using ridge regression. *Object* denotes the average R^2 over all object ground truth information and *Proprio* the ground truth proprioceptive state. *Action* predicts the action chunk from latent z_t ; parenthesized IDM scores use both z_t, z_{t+K} where reported. *Eff. rank* denotes effective rank (Roy & Vetterli, 2007).

Environment	Model	SIGReg	Eff. rank	Object $\uparrow$	Proprio $\uparrow$	Action (IDM) $\uparrow$
Push-T	LeWAM	Yes	42.90	0.90	0.86	0.40(0.75)
		No	10.94	0.88	0.85	0.38(0.73)
	LeWAM (MSE)	No	6.96	0.66	0.86	0.36 (0.74)
	LeWM	Yes	94.20	0.95	0.80	0.29(0.71)
		No	16.23	0.00	0.00	0.00 (0.00)

Why Train the Visual Encoder End-to-End? Frozen DINOv2 (Oquab et al., 2023) features underperform end-to-end features in our setup (Figure 8(b)). Across the four tasks, direct success averages 86.7% with the jointly trained encoder, versus 79.0% with frozen CLS features and 75.8% with frozen patch features. Joint training thus improves the representation used for action selection; the additional gain from planning shows that its learned dynamics also support action refinement. Appendix C.4 provides aligned visual reconstructions and recorded-action open-loop rollouts for four simulated tasks.

Why Model Stochastic Action Distributions? With the same latent-dynamics objective, flow matching yields larger planning gains than an MSE head whose point predictions are perturbed with Gaussian noise (Figure 8(c)). Without SIGReg, flow matching also retains more linearly decodable object information ($R^2 = 0.88$ versus 0.66), with comparable proprioception scores (0.85 versus 0.86). Deterministic MSE targets the conditional mean $\mathbb{E}[A_{t,C} \mid z_{\leq t}, c_t]$, whereas flow matching models the conditional action distribution. It can therefore retain distinctions between states whose action distributions differ despite similar means. Appendix C.5 illustrates this distinction in a two-door navigation experiment, where LeWAM presents a explicit multi-modal action distribution.

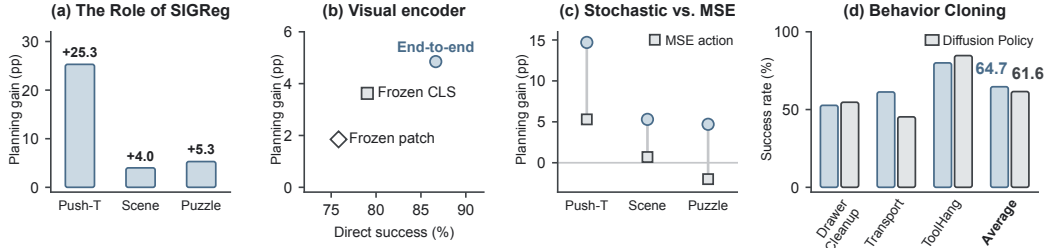


Figure 8: (a) Planning gain (percentage points) from SIGReg. (b) Direct success and planning gain for visual encoders, averaged over four tasks. (c) Planning gain for flow (circles) and MSE (squares) action heads. (d) Task-completion success for LeWAM and Diffusion Policy.

Isolating Action-Conditioned Dynamics. To examine the dynamics learned under LeWAM’s joint objective, we remove its action prediction head at evaluation time and use CEM to plan with only its action-conditioned latent dynamics. Compared with LeWM, which trains action-conditioned dynamics without policy learning, LeWAM produces comparable visual predictions and achieves competitive planning success on Push-T. On Cube, a 3D manipulation task, LeWAM has clearer visual predictions, especially in control-relevant feature, e.g., gripper and object, and higher planning success (Appendix C.4, Figure 13, and Table 10). These results show that joint action learning is compatible with world modeling and even improve the control-relevant parts.

4.4 BEHAVIOR CLONING AND PHYSICAL ROBOT CONTROL

While visual planning relies on goal images to optimize actions online, goal-agnostic tasks require executing behaviors directly. In this section, the robot must implicitly infer task progress and react to changing object poses and contacts without explicit goal conditioning. We train LeWAM without

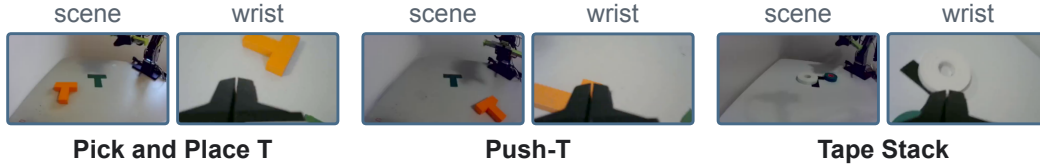


Figure 9: Paired scene and wrist view examples from three physical-robot tasks.

goal context c_t while retaining other objectives. At test time, the policy directly samples action chunks from observation history for execution, bypassing the test-time planning.

Task Completion in Simulation. We benchmark LeWAM on three simulated tasks requiring complex manipulation without goal prompts: DrawerCleanup, Transport, and ToolHang. Figure 8(d) compares task-completion success with Diffusion Policy (Chi et al., 2023). LeWAM achieves competitive performance on two tasks and improves success by 16 percentage points on Transport. These results show that LeWAM supports goal-agnostic behavior cloning alongside goal-conditioned planning. Per-task standard deviations appear in Appendix Table 11.

Physical-Robot Control. Several prior JEPAs, such as DINO-WM (Zhou et al., 2024), PLDM (Sobal et al., 2025), and LeWM (Maes et al., 2026b), evaluate control only in simulation, while V-JEPA 2-AC (Assran et al., 2025) plans on physical robots over a frozen pretrained encoder. We evaluate LeWAM on a real robotic platform through direct policy execution, testing whether an end-to-end JEPA can handle real-world contact dynamics and visual variation in closed-loop control.

We evaluate both the end-to-end LeWAM and a variant that supplies pretrained features from a frozen V-JEPA 2.1 (Mur-Labadia et al., 2026) backbone on Pick and Place T, Push-T, and Tape Stack. The physical-robot baselines are Diffusion Policy (Chi et al., 2023) and $\pi_{0.5}$ (Physical Intelligence et al., 2025), a vision-language-action policy. All methods are trained or finetuned on our collected teleoperation data and directly execute the first half of each predicted action chunk during inference. Implementation details appear in Appendix D.

Table 2: Physical robot evaluation. Successful trials out of 30 on Pick and Place T, Push-T, and Tape Stack. GFLOPs count one inference for a full action chunk, including all denoising steps. One full LeWAM action chunk uses 270.2 GFLOPs versus 3603.6 GFLOPs for $\pi_{0.5}$. Figure 1(d) expresses the reciprocal as action chunks per TFLOP, a $13.3\times$ difference in operation-count efficiency.

Method	Pick and Place T	Push-T	Tape Stack	Average	GFLOPs
Diffusion Policy (Chi et al., 2023)	19/30	16/30	0/30	38.9%	270.0
$\pi_{0.5}$ (Physical Intelligence et al., 2025)	14/30	3/30	9/30	28.9%	3603.6
LeWAM (Frozen V-JEPA 2.1 Encoder)	15/30	12/30	5/30	35.6%	270.2
LeWAM	16/30	8/30	6/30	33.3%	270.2

Table 2 shows that LeWAM is competitive with both baselines. With the same demonstrations, its encoder trained from scratch achieves a similar average success rate to the frozen V-JEPA 2.1 variant. Joint prediction of latent states and actions thus supports closed-loop physical control from teleoperation demonstrations.

On held-out episodes, both LeWAM variants reconstruct the manipulated object and coarse robot motion, with less fine detail in decoded rollouts (Appendix Figure 16).

5 CONCLUSION AND LIMITATIONS

In this work we introduce LeWAM, an end-to-end JEPA for joint prediction of future latent states and actions, that supports control through both planning and policy execution. On goal-reaching environments, LeWAM demonstrates improved performance in dynamics-based planning over previous JEPA-based world models. On behavioral cloning from expert demonstrations on both simulated and physical robots, LeWAM is competitive with established policy baselines.

Limitations and Future Work.

- **Data Coverage.** Our task suite combines expert demonstrations and non-expert interactions at limited scale. Broader play data and exploration on contact-rich tasks would extend coverage of difficult contact transitions and support training across a wider range of behaviors.
- **Generalization.** Learning action-conditioned dynamics could support broader generalization than policy-only approaches, motivating tests on unfamiliar objects, camera viewpoints, and contact states.
- **Single-Task and Single-Embodiment Learning.** Models are trained or adapted separately for each task and embodiment. Multitask training across robots and language-conditioned tasks would extend the shared dynamics and policy to broader settings.
- **Integration with Hierarchical Planning.** LeWAM provides low-level planning and control. A high-level planner could decompose longer tasks into subgoals, each executed through LeWAM’s continuous control.
- **Scalability of Training & Planning.** The visual-planning model has approximately 17M parameters. Larger models and datasets would extend training scale, while larger candidate populations and more refinement steps would extend test-time search.

REPRODUCIBILITY STATEMENT

The experiments in Section 4 use the architectures and baseline configurations in Appendix A, with datasets and evaluation protocols in Appendix B. The project website and code repository are linked on the first page.

ETHICS STATEMENT

The experiments collect no new human-subject data. Reused environments, datasets, models, and software remain subject to their original licenses. Physical deployment (Section 4.4) requires independent safety mechanisms.

AI USE STATEMENT

LLM-based assistants helped with manuscript polishing and language refinement. The authors designed and conducted the research, reviewed the analyses and manuscript, and take responsibility for its content.

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Appendix

Supplementary material for **LeWAM: End-to-End World and Action Modeling with JEPAs**

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A IMPLEMENTATION DETAILS

Temporal Sampling. Given observation-history stride Δ , the predictor in Section 3.1 uses two history latents $\mathbf{z}_{\leq t} = (\mathbf{z}_{t-\Delta}, \mathbf{z}_t)$, a dynamics stride of $K = 5$ raw steps, and an action chunk of $C = 2K = 10$ raw steps. We use a stride of $\Delta = K$ for goal-reaching tasks and $\Delta = 1$ for task-completion frames. To predict $\mathbf{z}_{t+5}$, the dynamics branch receives all five actions $\mathbf{a}_t, \dots, \mathbf{a}_{t+4}$, and the flow policy models all ten actions $\mathbf{a}_t, \dots, \mathbf{a}_{t+9}$. Only frames are subsampled. During imagined rollouts, the history update retains the two most recent latent states.

Inner Architecture. The observation encoder is a ResNet-18-style network (He et al., 2016) trained from scratch. Its convolutional backbone uses GroupNorm with 16 groups. Input images are cropped from 224×224 to 202×202 , with random offsets during training and a center crop at evaluation. A 1×1 convolution produces 32 spatial maps, and a spatial softmax pooling converts these to 32 expected normalized image-coordinate pairs. After that, a two-layer projection MLP with BatchNorm maps these 64 features through a hidden width of 384 to a 192-dimensional latent state, which is the same as LeWM (Maes et al., 2026b) in simulations for a fair comparison. The MoT predictor has width 192, 4 blocks, 4 attention heads, and dropout 0.1. Its token sequence contains two history latents, one learned state query, five clean action tokens, and ten action flow slots. Learned type and positional embeddings distinguish history, state query, clean action, and action flow tokens. The state readout uses LayerNorm followed by a linear map to the 192-dimensional latent, while the action readout maps each action flow slot to a velocity vector in raw-action space.

Goal information $\mathbf{c}_t$ is optional and is omitted for goal-free task completion. For goal-conditioned visual planning, $\mathbf{c}_t$ contains the goal image latent $\mathbf{z}_g$ and the normalized remaining horizon. The goal latent is supplied as an additional token in the action branch, while only noisy action tokens can attend to it. The horizon h between the current timestep and the goal is normalized into a $[0, 1]$ range relative to a maximal value $H_{\max}$ which dictates the longest distance-from-goal seen during training. h modulates the noisy action tokens using AdaLN-Zero (Peebles & Xie, 2023) blocks with hidden width 512 and a 128-dimensional sinusoidal embedding.

In total, the model has approximately 16.9M trainable parameters in visual planning. For the simulated experiments, action sampling uses 8 Euler steps to balance speed and quality. The reported MSE dynamics configuration uses an online encoder target, no state noise, no state flow integration, no momentum target encoder, only SIGReg with 1,024 projections and 17 knots. For the training objective in Equation 5, we set $\lambda_{\text{sig}} = 0.04$.

Training Recipe in Simulated Environments. For goal-conditioned visual planning, LeWAM runs use a batch size of 128 and train for 50 epochs, with a base learning rate of 1.5×10^{-4} and five warmup epochs. The goal-free task-completion runs use a batch size of 64 and train for 120 epochs, with a base learning rate of 10^{-4} and ten warmup epochs. Both use AdamW with weight decay 10^{-4} and a uniform learning rate across the encoder and MoT predictor. The learning rate increases linearly during warmup and then follows an epochwise cosine schedule. Training uses full precision, a gradient-norm clip of 1, and seed 42. We split valid decision points into 90% training and 10% validation, then sample training points with replacement within each epoch. After each epoch, we retain the checkpoint with the lowest validation action flow-matching loss, where task success is not used for checkpoint selection.

Tokens and Conditioning in MoT. With the optional goal context $\mathbf{c}_t$ omitted, the MoT sequence is

$$\left[\mathbf{z}_{t-\Delta}, \mathbf{z}_t, \mathbf{q}_{t+K}^z, \mathbf{A}_{t,K}, \mathbf{A}_{t,C}^{(s)} \right]. \quad (9)$$

Here $\mathbf{A}_{t,K}$ and $\mathbf{A}_{t,C}^{(s)}$ expand into K clean and $2K$ noisy action tokens, respectively. The first three tokens are processed by f_z , and both action groups by f_a . Each MoT branch has separate query, key, value, output, normalization, and feedforward parameters, while one global masked self-attention operates over the complete token sequence. Flow timestep s modulates only the noisy action slots through AdaLN-Zero.

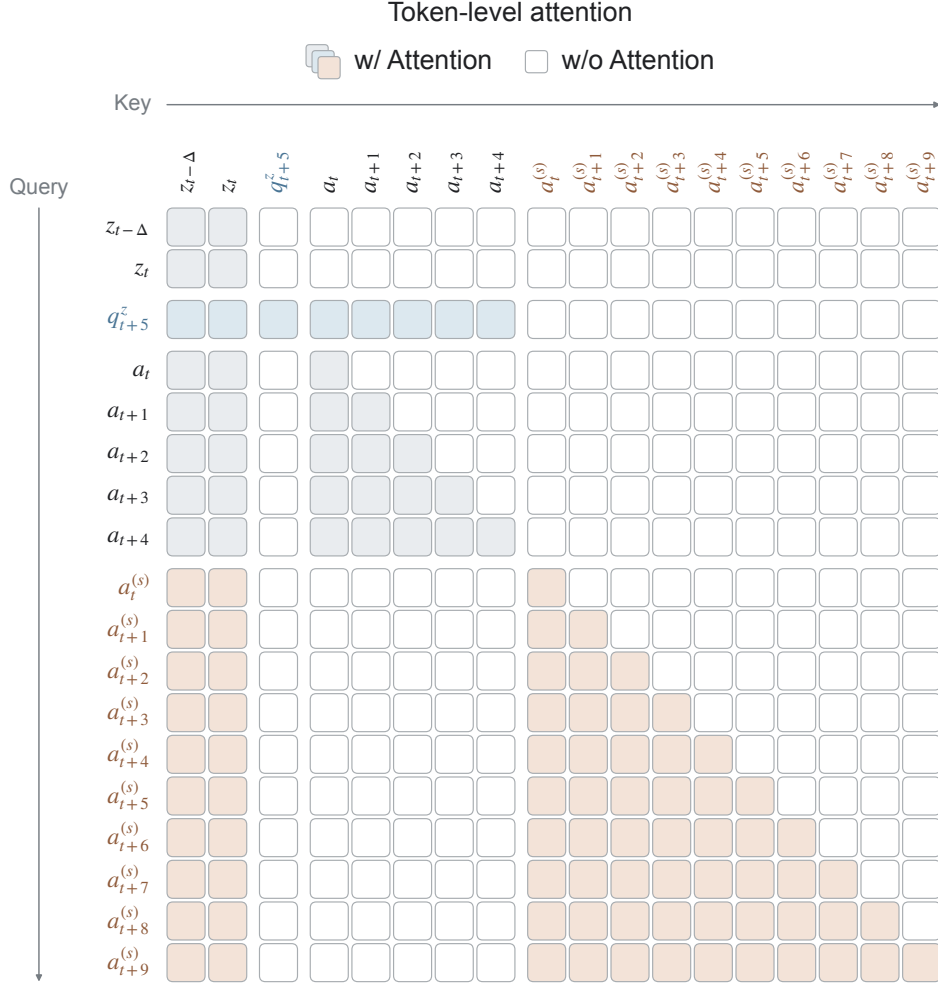


Figure 10: **Token-Level MoT Attention.** Expansion of Figure 3 for two history latents, one learned state query, five clean actions, and ten interpolated action tokens. Rows are queries; columns are keys. Filled cells permit attention; white cells are masked. Blue and brown mark latent state and action velocity prediction queries; gray marks context. Wider gaps separate token groups. Optional goal conditioning c_t is omitted. Δ is the history stride, and action indices remain at the raw environment rate.

Attention Mask. Let $\mathcal{V}(x)$ denote the tokens visible to query x . The mask below describes the sequence without optional goal context c_t . When this context is provided, the goal token is additionally visible to the noisy action tokens only. Action prefixes begin at the current chunk’s start t ; padded history keys are excluded:

$$\begin{aligned}
 \mathcal{V}(z_i) &= \{z_{t-\Delta}, z_t\}, & i &\in \{t-\Delta, t\}, \\
 \mathcal{V}(q_{t+K}^z) &= \{z_{t-\Delta}, z_t, q_{t+K}^z, \mathbf{A}_{t,K}\}, \\
 \mathcal{V}(a_j) &= \{z_{t-\Delta}, z_t, \mathbf{a}_{\leq j}\}, & j &\in \{t, \dots, t+K-1\}, \\
 \mathcal{V}(a_j^{(s)}) &= \{z_{t-\Delta}, z_t, \mathbf{a}_{\leq j}^{(s)}\}, & j &\in \{t, \dots, t+2K-1\}.
 \end{aligned} \tag{10}$$

Clean action tokens therefore cannot relay target information to history or action flow tokens across layers. A visualization of the attention mask is provided in Figure 10.

A.1 LEWAM PLANNING

Optimization Details. The goal cost $\mathcal{C}$ is defined in Eq. (7). During refinement, model parameters remain frozen and only the normalized action plans receive gradients. For each sampled plan $\mathbf{A}^{\text{sample}}$, Adam minimizes the goal cost with a quadratic proximity penalty:

$$\mathcal{J}_{\text{prox}}(\mathbf{A}; \mathbf{A}^{\text{sample}}) = \mathcal{C}(\mathbf{A}) + \lambda_{\text{prox}} \|\mathbf{A} - \mathbf{A}^{\text{sample}}\|_F^2. \quad (11)$$

The penalty affects the updates, while candidate retention and final selection use the unpenalized goal cost $\mathcal{C}$, and we clip normalized actions to $[-a_{\max}, a_{\max}]$ after each update. This penalty aims at discouraging the huge changes on actions, since the action flow already provide an effective prior.

Algorithm 1 Candidate refinement and selection for LeWAM planning

Require: $\mathbf{z}_{\leq t}, \mathbf{c}_t, \mathbf{z}_g, L, N, J, \eta, \lambda_{\text{prox}}, a_{\max}$
Sample candidates from the learned policy
1: $\mathbf{A}_{1:N}^{\text{sample}} \leftarrow N$ policy rollouts from $(\mathbf{z}_{\leq t}, \mathbf{c}_t)$ for L/K blocks
2: $c_{1:N} \leftarrow \mathcal{C}(\mathbf{A}_{1:N}^{\text{sample}})$
3: $\mathbf{A}_{1:N}^* \leftarrow \mathbf{A}_{1:N}^{\text{sample}}, \quad \mathbf{A}_{1:N} \leftarrow \mathbf{A}_{1:N}^{\text{sample}}$
4: $c_{1:N}^* \leftarrow c_{1:N}$
Refine each candidate
5: **for** $j = 0, \dots, J - 1$ **do**
6: $\mathbf{A}_{1:N} \leftarrow \text{AdamStep}(\mathbf{A}_{1:N}, \nabla_{\mathbf{A}_{1:N}} \mathcal{J}_{\text{prox}}(\mathbf{A}_{1:N}; \mathbf{A}_{1:N}^{\text{sample}}), \eta)$
7: $\mathbf{A}_{1:N} \leftarrow \text{clip}(\mathbf{A}_{1:N}, -a_{\max}, a_{\max})$
8: $c_{1:N} \leftarrow \mathcal{C}(\mathbf{A}_{1:N})$
9: $\mathbf{A}_n^* \leftarrow \mathbf{A}_n, \quad c_n^* \leftarrow c_n \quad \forall n \text{ such that } c_n < c_n^*$
10: **end for**
Select the lowest-cost iterate
11: $n^* \leftarrow \arg \min_{n=1, \dots, N} c_n^*$
12: **return** $\mathbf{A}_{n^*}^*$

We let the Adam optimizer use a learning rate of 0.05, gradient-norm clipping at 10, $a_{\max} = 3$, and $\lambda_{\text{prox}} = 0.01$.

A.2 BASELINE IMPLEMENTATIONS

The baselines in Section 4.1 use the released stable-worldmodel codebase (Maes et al., 2026a), retaining its environment wrappers, dataset interface, action bounds, and evaluation loop. Unless stated below, each method follows its official implementation and hyperparameters.¹ These configurations describe goal-conditioned comparisons. Methods share observations, actions, start and goal pairs, and success predicates. We list deviations from released configurations.

- **LeWM (IDM).** This is the GC-IDM (Nguyen et al., 2026) configuration: a pretrained LeWM checkpoint trained by us is frozen, then a goal-conditioned inverse-dynamics model is post-trained. We follow its official two-stage implementation. Goal reaching samples a remaining horizon uniformly from 1 to 50 and clips it to the episode boundary.
- **LeWM (CEM).** We follow the official LeWM training and CEM evaluation settings (Maes et al., 2026b). We train a checkpoint for each task and use these checkpoints in all quantitative experiments with LeWM as a baseline.
- **DINO-WM.** We evaluate the released patch-token model and a matched CLS-token variant; tokenization is their only architectural difference. The six added tasks use ten training epochs, while the original four retain their four-epoch checkpoints.
- **GCBC.** We use the goal-conditioned behavior-cloning baseline from the released codebase (Maes et al., 2026a), with its pretrained DINOv2 encoder frozen and the policy trained for 20 rather than

¹Official repositories: DINO-WM, stable-worldmodel, and GC-IDM.

100 epochs. Short-horizon goal reaching samples a uniformly later frame from the same episode. At evaluation, the policy executes its predicted five-step action chunks without planning.

Behavior-Cloning Baseline in Simulation. The task-completion comparison in Table 11 uses a policy trained on demonstrations from each task, conditioned on observation history without a goal image or goal-horizon input.

- **Diffusion Policy** (Chi et al., 2023). We use the convolutional variant (DP-C), with a ResNet-18 encoder, GroupNorm, and spatial softmax for each camera. The image features condition a one-dimensional U-Net with channel widths (512, 1024, 2048), kernel size five, and diffusion-step embedding dimension 128. Training minimizes noise-prediction MSE with a 100-step squared-cosine diffusion schedule; inference uses 100 DDPM steps. The policy takes two consecutive observations, predicts a 16-action sequence, and executes eight actions starting at the current timestep before resampling.

ToolHang uses the scene camera; DrawerCleanup and Transport use the scene camera and both wrist cameras. Inputs are 224×224 RGB images, with random 202×202 crops during training and center crops at evaluation, normalized to $[-1, 1]$. The policy uses visual observations only. Training windows advance at the raw frame rate, with boundary frames and actions repeated for padding. We randomly split windows into 90% training and 10% validation using seed 42. Each policy trains for 120 epochs with batch size 64 and AdamW, learning rate 10^{-4} , betas (0.95, 0.999), and weight decay 10^{-6} . The learning rate follows cosine decay after 500 warmup updates. Actions retain their recorded scale for ToolHang and Transport; DrawerCleanup uses min-max scaling to $[-1, 1]$. We select the checkpoint with the lowest validation denoising loss and evaluate its raw weights without exponential moving averaging. Each checkpoint is evaluated with seeds 0, 1, and 42, using 50 episodes per seed.

B BENCHMARKS AND EVALUATION

B.1 TASKS AND DATASETS

Task Selection. We retain the four environments used by LeWM (Maes et al., 2026b) for direct comparison, then add six tasks that broaden navigation and manipulation to larger mazes, interacting objects, coupled button states, precise tool placement, bimanual transfer, and articulated drawers (Figure 5). The ten tasks also span different action dimensions, observation views, trajectory lengths, and data-collection regimes. Together, they test the same model design under varied control and data conditions.

Data Provenance. We describe each dataset in the environment order of Figure 5.

- **Two-Room.** We use the dataset released with LeWM (Maes et al., 2026b), collected by a noisy heuristic navigation policy.
- **Reacher.** We use the dataset released with LeWM (Maes et al., 2026b), collected by a random policy with uniform distribution.
- **Push-T.** We use the non-expert interaction dataset released by DINO-WM (Zhou et al., 2024), generated by replaying the original expert trajectories with different levels of action noise.
- **Cube.** We use the dataset released with LeWM (Maes et al., 2026b) for the single-cube OGBench (Park et al., 2025) environment, collected by a scripted expert that picks and places the cube at sampled target locations. We just change the success criteria.
- **PointMaze-Large.** We use the official OGBench (Park et al., 2025) pointmaze-large-navigate-v0 dataset, collected by a noisy expert that navigates between randomly sampled locations.
- **Scene.** We collect data ourselves using the stable-worldmodel (Maes et al., 2026a) wrapper for OGBench (Park et al., 2025) Scene (swm/OGBScene-v0) and its ExpertPolicy, a noisy, closed-loop scripted oracle. The policy interacts with the cube, drawer, window, and buttons; we record a single camera view.
- **Puzzle.** We collect data ourselves using the stable-worldmodel (Maes et al., 2026a) wrapper for the 3×3 OGBench (Park et al., 2025) Puzzle environment and its scripted ExpertPolicy. The policy presses buttons in sampled sequences; we record a single camera view.

- **ToolHang.** We use the official human behavior-cloning demonstrations from robomimic (Mandlekar et al., 2022).
- **Transport.** We use the official generated behavior-cloning demonstrations from DexMimicGen (Jiang et al., 2025).
- **DrawerCleanup.** We use the official generated behavior-cloning demonstrations from DexMimicGen (Jiang et al., 2025).

The three behavior-cloning datasets broaden the visual-planning comparison to complex contact-rich behaviors with higher-dimensional actions, although their task-focused demonstrations may limit the diversity of sampled visual goals.

Table 3 summarizes all ten tasks, including observation views, raw action dimensions, and dataset scale. Recorded lengths describe the data, while Appendix B.2 specifies evaluation horizons and execution budgets. Recorded duration uses the native sampling rate before temporal subsampling; synchronized views of the same trajectories are counted once. **ToolHang** uses one oblique scene view (about 45°), while **Transport** and **DrawerCleanup** use a scene view and both wrist views.

Table 3: Simulation Benchmark Datasets. Ten tasks in Figure 5 order. Views gives camera-stream count; Dim. is the raw action dimension. Statistics cover complete source datasets before splits. Total hours are computed before rounding per-task values.

Task	Collection regime	Views	Dim.	Traj.	Mean frames	Total frames	Hz	Hours
Two-Room	Noisy heuristic	1	2	10,000	92	920,809	10	25.58
Reacher	Uniform random	1	2	10,000	201	2,010,000	25	22.33
Push-T	Non-expert	1	2	18,685	125	2,336,736	10	64.91
Cube	Scripted expert	1	5	10,000	201	2,010,000	20	27.92
PointMaze-Large	Scripted navigation	1	2	1,000	1,000	1,000,000	10	27.78
Scene	Scripted expert	1	5	10,024	200	2,004,800	20	27.84
Puzzle (3×3)	Scripted expert	1	5	10,024	200	2,004,800	20	27.84
ToolHang	Human demos	1	7	200	480	95,962	20	1.33
Transport	Generated demos	3	14	1,029	408	419,712	20	5.83
DrawerCleanup	Generated demos	3	24	1,026	291	298,235	20	4.14
Reported total (10)				71,988		13,101,054		235.51

B.2 GOALS AND SUCCESS CRITERIA

For the comparisons in Section 4.1, unless stated otherwise, if not specify, we report mean success and standard deviation over three seeds with 50 episodes per seed. Methods within a protocol share observations, action limits, execution budgets, and success predicates.

Goal-Conditioned Visual Planning. In Section 4.2, the initial state is sampled from an offline trajectory. Except on **Two-Room**, the frame 25 steps later supplies the visual goal, and each controller receives 50 environment actions. On **Two-Room**, we use a goal offset of 100 environment steps and an execution budget of 150 actions, following **LeWorldModel** (Maes et al., 2026b). We report first-hit success for comparison with the released **LeWM** evaluation (Maes et al., 2026b); the **Cube** predicate correction is detailed below. **Scene** and **Puzzle** follow the same protocol; their random-policy and zero-action controls appear alongside those for the other environments in Table 4.

Policy-Floor Controls. To ensure the benchmarks are reliable in reflecting the model ability, Table 4 also reports zero-action and random-policy success under the full-trajectory task-completion evaluation. These blind controls cover the benchmark environments.

Cube Success Predicate. Specifically, we add an end-effector (EEF) check to the **Cube** success criterion to exclude spurious successes. The original **Cube** criterion checks whether the cube center lies within 0.04 m of the target, regardless of the gripper pose. For 25-step goals, this credits zero action on 42.7% of episodes (Table 4). The corrected criterion additionally requires the end effector

Environment	Raw action dim.	25-step goal reaching		Task Completion	
		Zero-action	Random	Zero-action	Random
Two-Room	2	7.3 $\pm$ 0.9	25.3 $\pm$ 5.0	0.0 $\pm$ 0.0	2.7 $\pm$ 2.5
Reacher	2	3.3 $\pm$ 1.9	12.0 $\pm$ 1.6	0.0 $\pm$ 0.0	12.0 $\pm$ 4.9
Push-T	2	0.0 $\pm$ 0.0	1.3 $\pm$ 0.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Cube	5	42.7 $\pm$ 6.6	44.0 $\pm$ 1.6	4.7 $\pm$ 2.5	6.0 $\pm$ 1.6
Cube (+EEF)	5	2.7 $\pm$ 0.9	4.7 $\pm$ 0.9	4.7 $\pm$ 2.5	6.0 $\pm$ 1.6
PointMaze-Large	2	6.0 $\pm$ 0.0	9.3 $\pm$ 1.9	0.7 $\pm$ 0.9	6.0 $\pm$ 3.3
Scene	5	2.7 $\pm$ 2.3	2.7 $\pm$ 2.3	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Puzzle	5	0.7 $\pm$ 1.2	0.7 $\pm$ 1.2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
ToolHang	7	22.7 $\pm$ 2.5	26.0 $\pm$ 1.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Transport	14	3.3 $\pm$ 0.9	2.7 $\pm$ 0.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
DrawerCleanup	24	0.0 $\pm$ 0.0	6.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0

Darker shading indicates a higher control success rate.

Table 4: **Action Dimensions and Policy-Floor Controls.** Rows cover all ten environments in Figure 5, plus a Cube (+EEF) row isolating the corrected end-effector predicate. Raw action dimensionality is reported beside zero-action and random-policy success rates (%) for short-horizon goal reaching and full-trajectory task completion, averaged over three seeds. These blind controls measure baseline success from zero actions and random actions. Cube has the same task-completion values in both rows because the end-effector correction applies only to short-horizon goal reaching. DrawerCleanup exposes 24 raw coordinates; the policy predicts 22 active dimensions after removing two constant joints.

to lie within 0.04 m of its goal-frame position, reducing this floor to 2.7%. Figure 11 illustrates the object-only failure case.

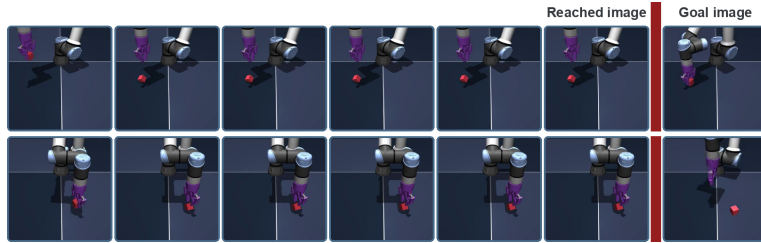


Figure 11: **A Degenerate Cube Success under Goal Reaching.** Each row shows a policy rollout (left, frames over time) beside its goal frame (far right, after the divider). Throughout the execution, the gripper never reaches the goal end-effector pose, yet the cube stays within the 0.04 m object-only threshold of the goal, so object-position-only predicate scores the episode a spurious success. Our end-effector term (Table 4) rejects these degenerate successes by additionally requiring the gripper to reach the goal pose.

B.3 PLANNING AND TIMING PROTOCOLS

Goal-Horizon Sweep. The sweep in Figure 7 and Table 5 starts from the first frame of each held-out trajectory and uses the recorded frames 25, 50, 75, or 100 steps later as goals. All tasks use the shared planning and replanning settings in Section 3.2.

Planner-Sensitivity Protocol. On Push-T, we vary Adam refinement updates while fixing the candidate count at 32, then vary the number of sampled candidates while fixing refinement at five updates. Goal-reaching success and per-plan latency are measured over three seeds with 50 episodes per seed. The measured curves appear in Figure 12 in Appendix C.1.

Timing Scope. Table 6 in Appendix C.1 reports time per planner call. Figure 1(c) averages the per-task throughput across its four tasks, using 1000/milliseconds per plan to obtain plans/s. The compared settings are 300×30 CEM for LeWM (Maes et al., 2026b) and five gradient steps for LeWAM. All compared neural networks use the same H100 GPU.

C ADDITIONAL RESULTS

C.1 VISUAL PLANNING RESULTS

Table 5 gives the measured goal-horizon sweeps behind Figure 7. Table 6 gives the per-call timing behind Figure 1(c). Appendix B.3 specifies the evaluation and timing protocols.

Table 5: **Goal-Reaching Success across Horizons.** Success rate (%), mean $\pm$ standard deviation. Measured values for Figure 7. The Cube horizon sweep is distinct from the sampled-start 25-step evaluation in Figure 6.

Task	Method	Goal horizon (steps)			
		25	50	75	100
Push-T	LeWAM	94.7 ± 3.1	60.7 ± 3.1	44.0 ± 3.5	43.3 ± 4.6
	LeWM (CEM)	68.7 ± 3.1	12.0 ± 4.0	3.3 ± 2.3	2.7 ± 1.2
Cube	LeWAM	100.0 ± 0.0	88.0 ± 2.0	80.7 ± 2.3	82.0 ± 2.0
	LeWM (CEM)	100.0 ± 0.0	0.0 ± 0.0	4.0 ± 3.5	4.0 ± 3.5
Scene	LeWAM	82.7 ± 3.1	73.3 ± 8.1	65.3 ± 3.1	56.0 ± 4.0
	LeWM (CEM)	44.7 ± 13.0	31.3 ± 2.3	37.3 ± 5.8	36.0 ± 6.0
Puzzle	LeWAM	74.0 ± 12.5	14.0 ± 3.5	4.0 ± 3.5	3.3 ± 1.2
	LeWM (CEM)	10.0 ± 0.0	4.7 ± 2.3	4.0 ± 2.0	0.7 ± 1.2

Table 6: **Planning Speed across Four Visual-Goal Tasks.** Each plan is one planner call. LeWM (CEM) (Maes et al., 2026b) uses 300×30 CEM; LeWAM uses five gradient steps. Action dimensions are raw environment action dimensions. Throughput is the reciprocal of time per plan. The mean row averages per-task throughput, and its ratio compares those two means.

Task	Action dim.	LeWM (CEM)		LeWAM		Speedup
		ms/plan	plans/s	ms/plan	plans/s	
Scene	5	705	1.42	18.8	53.19	$37.5\times$
Puzzle	5	731	1.37	16.3	61.35	$44.8\times$
Cube	5	771	1.30	21.1	47.39	$36.5\times$
Push-T	2	414	2.42	20.8	48.08	$19.9\times$
Task mean	—	—	1.62	—	52.50	$32.3\times$

Planner Sensitivity. In the Push-T sweep in Figure 12, success plateaus after five refinement updates while latency continues to rise. Increasing the number of sampled candidates beyond eight gives no consistent success gain in this sweep.

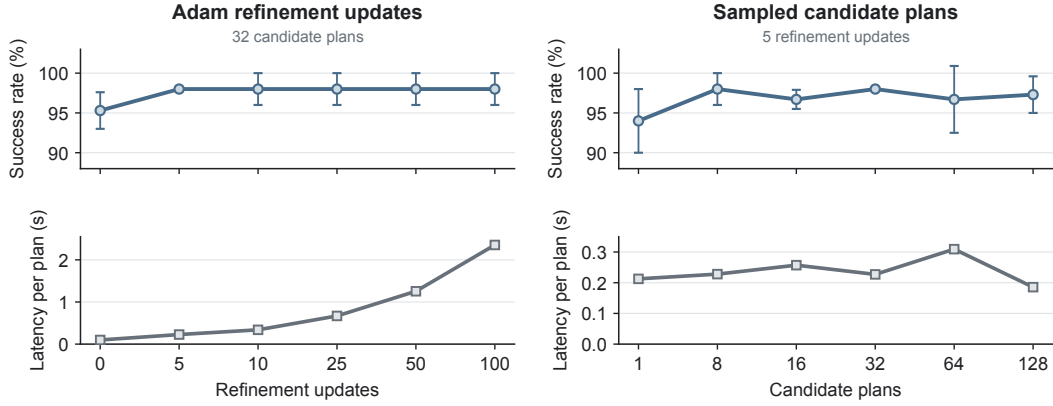


Figure 12: **Planner Sensitivity on Push-T.** Goal-reaching success (mean and standard deviation across three seeds, 50 episodes each) and per-plan latency versus (left) Adam refinement updates with 32 candidate plans and (right) sampled candidate plans with five refinement updates.

C.2 COMPONENT AND VISUAL-ENCODER ABLATIONS

Table 7 records the SIGReg and action-head comparisons in Figure 8(a,c). All variants use epoch-30 checkpoints. Table 8 reports the separate encoder comparison in Figure 8(b). Direct control executes the goal-conditioned policy without search. Planning performs test-time action search through the learned dynamics. The encoder panel reports equal-weight averages over Push-T, Cube, Scene, and Puzzle.

Table 7: **SIGReg and Action-Head Ablations.** Direct and planning success rates (%) using epoch-30 checkpoints for all variants. Gain is planning minus direct success in percentage points. Standard deviations are reported where available; none is inferred for gains.

Environment	Variant	Direct	Planning	Gain (pp)
Push-T	LeWAM	82.7 ± 8.1	97.3 ± 3.1	+14.7
	w/o SIGReg	69.3 ± 10.1	72.0 ± 3.5	+2.7
	MSE action head	90.0 ± 4.0	95.3 ± 1.2	+5.3
Scene	LeWAM	92.0 ± 5.3	97.3 ± 1.2	+5.3
	w/o SIGReg	91.3 ± 2.3	93.3 ± 3.1	+2.0
	MSE action head	89.3 ± 4.2	90.0 ± 2.0	+0.7
Puzzle	LeWAM	76.0 ± 7.2	80.7 ± 11.0	+4.7
	w/o SIGReg	67.3 ± 9.5	75.3 ± 9.0	+8.0
	MSE action head	76.7 ± 6.1	74.7 ± 6.1	-2.0
Cube	LeWAM	99.3 ± 1.2	100.0	+0.7
	w/o SIGReg	99.3 ± 1.2	99.3 ± 1.2	0.0
	MSE action head	98.0 ± 2.0	99.3 ± 1.2	+1.3

Table 8: **Visual Encoders for Direct Control and Planning.** Success rate (%) with the same goal-conditioned evaluation protocol for all encoder variants. Direct executes the policy without search; planning uses test-time action search. Gain is planning minus direct success in percentage points. Averages are equal-weight means over the four tasks; gains use unrounded means.

Visual encoder	Mode	Push-T	Cube	Scene	Puzzle	Average
LeWAM (end-to-end)	Direct	85.3 $\pm$ 5.0	100.0 $\pm$ 0.0	88.0 $\pm$ 6.9	73.3 $\pm$ 5.8	86.7
	Planning	98.7 $\pm$ 1.2	100.0 $\pm$ 0.0	93.3 $\pm$ 3.1	74.0 $\pm$ 12.5	91.5
	Gain (pp)	+13.4	0.0	+5.3	+0.7	+4.9
frozen DINOv2 CLS	Direct	72.7 $\pm$ 7.6	98.7 $\pm$ 1.2	82.0 $\pm$ 4.0	62.7 $\pm$ 7.0	79.0
	Planning	78.7 $\pm$ 6.4	99.3 $\pm$ 1.2	91.3 $\pm$ 1.2	61.3 $\pm$ 13.0	82.7
	Gain (pp)	+6.0	+0.6	+9.3	-1.4	+3.6
frozen DINOv2 patch mean	Direct	66.0 $\pm$ 12.5	93.3 $\pm$ 2.3	84.0 $\pm$ 2.0	60.0 $\pm$ 7.2	75.8
	Planning	67.3 $\pm$ 8.1	98.7 $\pm$ 1.2	86.0 $\pm$ 4.0	58.7 $\pm$ 7.6	77.7
	Gain (pp)	+1.3	+5.4	+2.0	-1.3	+1.9

C.3 PROBING LATENT REPRESENTATIONS

Table 9 extends the probing results presented on Push-T in Section 4.3 to Cube, Scene, Puzzle. All probes report R^2 from ridge regression on a 4000/1000-frame train/test split. *Object* is the mean R^2 over object ground-truth variables and *Proprio* over proprioceptive variables. *Action (IDM)* predicts the action chunk from z_t (and from $[z_t, z_{t+K}]$, in parentheses where reported). *Eff. rank* is the effective rank measure introduced in (Roy & Vetterli, 2007).

Table 9: **Effective Rank and Linear Probing.** Results on Push-T, Cube, Scene, and Puzzle, with the Object, Proprio, and Action columns reporting linear-probe R^2 . Rows compare models with and without SIGReg; parenthesized values use both z_t and z_{t+K} for inverse-dynamics probing.

Environment	Model	SIGReg	Eff. rank	Object $\uparrow$	Proprio $\uparrow$	Action (IDM) $\uparrow$
Push-T	LeWAM	Yes	42.90	0.90	0.86	0.40 (0.75)
		No	10.94	0.88	0.85	0.38 (0.73)
	LeWAM (MSE)	No	6.96	0.66	0.86	0.36 (0.74)
	LeWM	Yes	94.20	0.95	0.80	0.29 (0.71)
		No	16.23	0.00	0.00	0.00 (0.00)
Cube	LeWAM	Yes	46.60	0.99	0.67	0.37 (0.66)
		No	8.00	0.99	0.70	0.37 (0.68)
	LeWM	Yes	108.10	0.99	0.58	0.17 (0.55)
Scene	LeWAM	Yes	48.90	0.48	0.78	0.39 (0.74)
		No	12.50	0.40	0.77	0.43 (0.74)
	LeWM	Yes	111.10	0.40	0.37	0.07 (0.35)
Puzzle	LeWAM	Yes	46.50	0.37	0.70	0.33 (0.69)
		No	5.80	0.22	0.66	0.31 (0.68)
	LeWM	Yes	97.40	0.60	0.42	0.16 (0.48)

C.4 VISUAL RECONSTRUCTIONS AND OPEN-LOOP ROLLOUTS

Figures 13 and 14 align recorded frames with pixel-decoded representations and latent predictions on Push-T, Cube, Scene, and Puzzle. Framewise reconstructions encode and decode each recorded observation independently. Open-loop predictions advance in five-step blocks from the initial observation context under fixed recorded actions. Each model receives the initial observed context and the recorded action sequence. We display the 51-frame clips at five-frame intervals and select the lowest-ID episode from the three recorded episodes available for each task.

The pixel decoders are trained post hoc on frozen world models using 150,000 sampled frames for 20 epochs; the displayed episodes are excluded from decoder training. These diagnostic clips use 50-epoch LeWAM checkpoints on Cube, Scene, and Puzzle, and an epoch-16 checkpoint on the full Push-T dataset. LeWM uses our 15-epoch checkpoints from its official implementation on the three 3D tasks and its released weights on Push-T. Each model uses its own encoder, training schedule, and post-hoc pixel decoder.

On Push-T, both models reproduce pusher motion and block translation and rotation under the recorded actions (Figure 13). On Cube, Scene, and Puzzle, LeWAM preserves clearer robot-arm, gripper, and object details in its reconstructions and decoded predictions (Figures 13 and 14). These sequences visualize the physical-state information available to each model’s learned dynamics. LeWAM has approximately 16.9M parameters, and LeWM (Maes et al., 2026b) has 15M.

Planning with Learned Dynamics. We remove LeWAM’s action prediction head at evaluation time and use CEM (Rubinstein, 1999) to plan with its action-conditioned latent dynamics. Table 10 reports the resulting goal-reaching success rates alongside LeWM (Maes et al., 2026b).

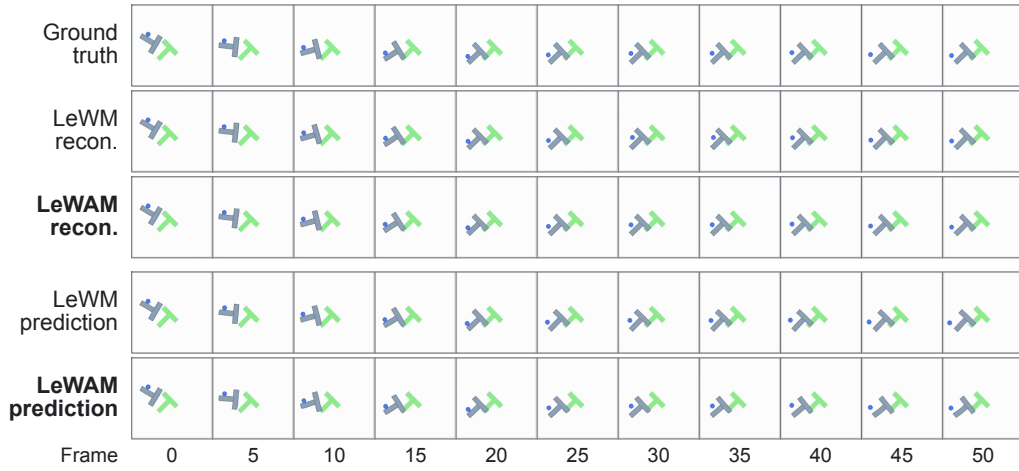
LeWAM achieves competitive success on Push-T and exceeds LeWM by 22.2 percentage points on Cube. Together with the visual reconstructions and recorded-action predictions in Figure 13, these results show that its learned action-conditioned dynamics support planning, including contact-rich 3D manipulation.

Table 10: **CEM Planning with Learned Dynamics.** Goal-reaching success rates (%), reported as mean $\pm$ standard deviation. LeWAM is evaluated with its action prediction head removed.

Model	Push-T	Cube
LeWM (CEM)	92.4 $\pm$ 1.2	63.8 $\pm$ 2.2
LeWAM (CEM)	84.0 $\pm$ 1.3	86.0 $\pm$ 3.2

(a) Push-T

Episode 1235



(b) Cube

Episode 638

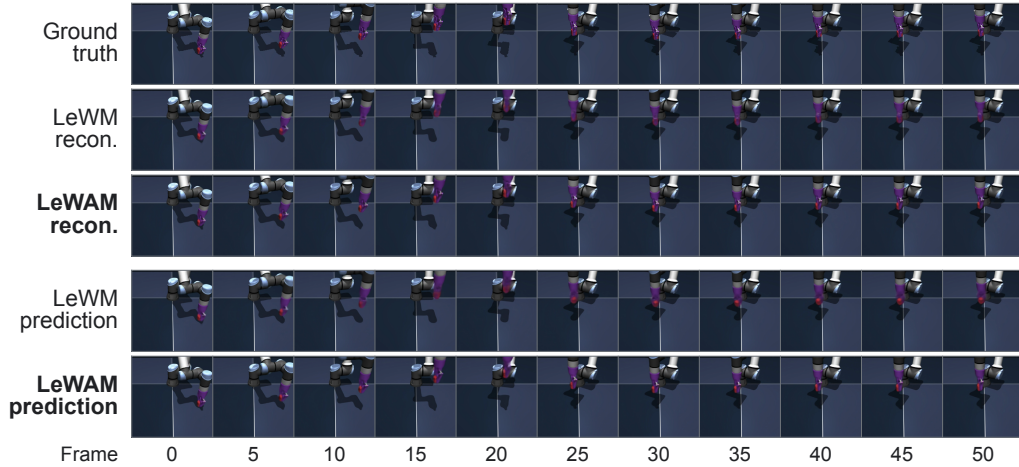


Figure 13: **Aligned Visual Sequences on Push-T and Cube.** Each task shows recorded ground truth, framewise reconstructions for LeWM and LeWAM, and decoded open-loop predictions for both models. Columns use the same relative frame indices; predictions follow the recorded actions.

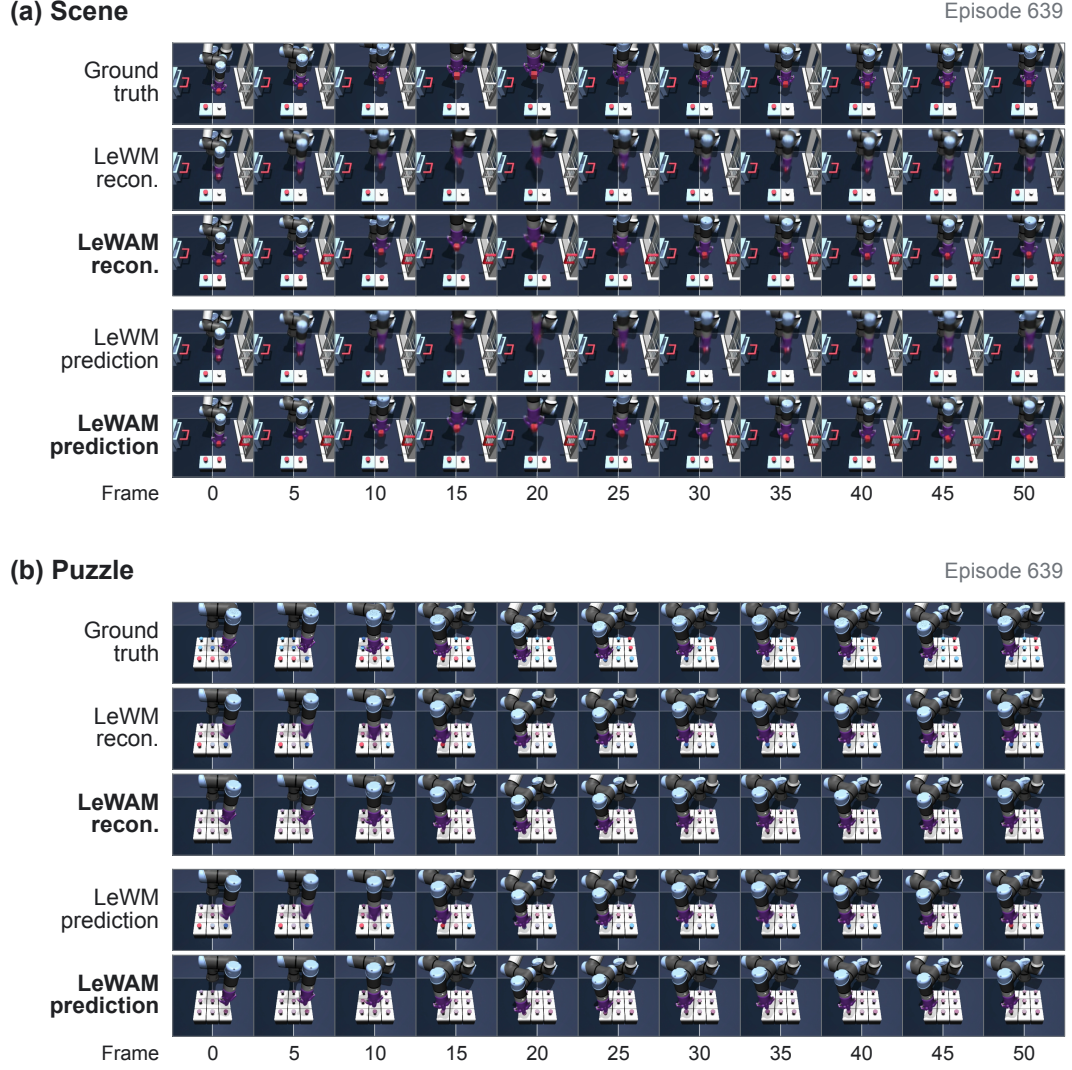


Figure 14: **Aligned Visual Sequences on Scene and Puzzle.** Rows show recorded ground truth, LeWM and LeWAM framewise reconstructions, and their decoded open-loop predictions. Columns use the same relative frame indices and the same recorded action sequence for both prediction rows.

C.5 MULTIMODAL ACTION DISTRIBUTIONS

To evaluate multi-route action distribution modeling, we collect expert demonstrations on a two-door navigation task with a fixed start and goal. Half of the expert trajectories pass through the upper door and half through the lower door, providing two valid routes between the same endpoints. We train LeWAM using (1) our default flow-matching action head and (2) a deterministic MSE action head on the same data. During closed-loop policy rollouts (Figure 15), the flow head follows both routes, whereas the deterministic MSE head concentrates on the lower route and produces some wall collisions. The flow head preserves both route choices present in the demonstrations, illustrating how a stochastic action model can supply distinct proposals when a goal admits multiple routes.

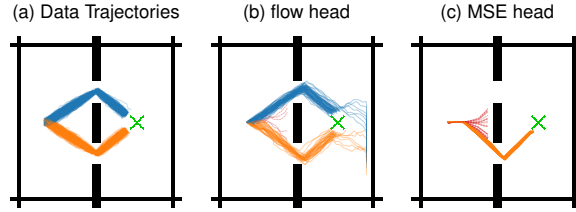


Figure 15: **Two-Door Demonstrations and Action-Head Rollouts.** Expert demonstrations (a), executed trajectories from the flow-matching action head (b), and executed trajectories from the deterministic MSE action head (c). Black segments mark walls, and the green cross marks the goal. Blue and orange indicate upper- and lower-door trajectories; dashed red paths end in collision.

C.6 SIMULATION TASK-COMPLETION RESULTS

Table 11 reports the per-task results and deviations behind Figure 8(d).

Table 11: Task completion. Mean success rates (%) $\pm$ standard deviation across three evaluation seeds, with 50 episodes per seed. Bold marks the better result.

Method	DrawerCleanup	Transport	ToolHang	Average
Diffusion Policy (Chi et al., 2023)	54.7 $\pm$ 1.2	45.3 $\pm$ 9.0	84.7 $\pm$ 3.1	61.6
LeWAM	52.7 $\pm$ 5.0	61.3 $\pm$ 9.9	80.0 $\pm$ 3.5	64.7

D PHYSICAL-ROBOT SETUP AND EVALUATION

This section documents the physical-robot experiments in Section 4.4.

D.1 PLATFORM AND DATA COLLECTION

Demonstrations are collected by joint-space teleoperation from a reBot Arm 102 leader to a reBot B601-DM follower through LeRobot (Cadene et al., 2026). The follower uses impedance control with gravity feedforward. One proficient operator collects successful demonstrations without intentional exploration. Between episodes, both arms return to the calibration zero pose before recording resumes.

A ZED-Mini records the scene at 672×376 pixels per eye, and a wrist-mounted Logitech C270 records at 640×360 pixels. The model uses the left scene view and wrist view, both cropped to 256×256 pixels; the right stereo view is recorded but unused. At 30 Hz, a Raspberry Pi 5 stores the three video streams, seven-dimensional joint targets as actions, and follower position, torque, and velocity as a 21-dimensional state.

Table 12: Datasets for the physical-robot tasks in Section 4.4.

Task	Episodes	Frames	Duration	Task Description
Pick and Place T	200	52,297	29 min	"Pick up the T block and place it onto the outline."
Push-T	100	61,996	34 min	"Push the T block onto the outline."
Tape Stack	100	23,726	13 min	"Stack the green roll of tape on the white roll of tape."

D.2 MODEL IMPLEMENTATION

The LeWAM models in Section 4.4 use the state MSE and action flow matching objectives from Section 3.1. In the frozen V-JEPA 2.1 variant, attentive pooling and a BatchNorm MLP reduce each camera frame to one summary token. Proprioceptive state and previous actions are linearly projected from raw vectors to the model dimension. Learned queries initialize the future-camera and action streams.

The real world LeWAM variants take two video context frames and predict three future frames, all sampled at 2.5 Hz. They use model width 512 and depth eight, and train with AdamW at batch size 256, using gradient accumulation when needed. Both variants use a ViT-B visual backbone. The end-to-end model trains its visual encoder from scratch; the frozen variant initializes its encoder from V-JEPA 2.1 (Mur-Labadia et al., 2026) and keeps it fixed. The action chunk length and execution interval are specified in Appendix D.3.

For Table 2, batch-1 GFLOPs include all inference steps for one full action chunk: 100 DDPM steps for Diffusion Policy (Chi et al., 2023), ten Euler steps for $\pi_{0.5}$ (Physical Intelligence et al., 2025), and ten Euler steps for each physical-robot LeWAM variant.

D.3 PHYSICAL-ROBOT BASELINE CONFIGURATIONS

For the comparison in Section 4.4, both baselines use the LeRobot implementations (Cadene et al., 2026), adapted to the scene and wrist cameras and seven-dimensional joint-target actions. Each policy returns 36 future actions and executes the first 18 at 30 Hz before requesting a new chunk, matching the LeWAM variants. The returned and executed chunks span 1.2 and 0.6 seconds, respectively. The demonstrations and evaluation procedure are described in Appendices D.1 and D.4, with 30 trials per task and method.

The `Pick and Place T` and `Push-T` training configurations resize both camera streams to 256×256 and retain seven joint torques from the recorded proprioceptive state. Actions and torques are normalized by mapping their training-set first and 99th percentiles to -1 and 1 , respectively. Training examples start every three recorded frames; the observation offsets and action targets retain their 30 Hz timing. We hold out 10% of episodes using split seed zero. The configurations specify 20,000 optimizer updates with effective batch size 256, a 500-update linear learning-rate warmup, and cosine decay. Validation runs every 200 updates; the trainer saves both the final checkpoint and the checkpoint with the lowest validation loss.

- **Diffusion Policy** (Chi et al., 2023). The robot implementation uses a separate ImageNet-initialized ResNet-18 with spatial softmax for each camera and a conditional one-dimensional U-Net with channel widths (336, 672, 1344). It conditions on two consecutive image and torque observations. Images are randomly cropped to 243×243 during training and center-cropped at evaluation. The internal diffusion horizon is 40 action positions, aligned from one step before the current observation; the policy returns positions corresponding to the next 36 actions. This 40-step internal sequence satisfies the U-Net’s temporal downsampling constraint. Training uses noise-prediction MSE with a squared-cosine schedule, and inference uses 100 DDPM steps. AdamW uses learning rate 10^{-4} , betas (0.95, 0.999), and weight decay 10^{-6} . Exponential moving averaging with decay 0.999 supplies the weights used for validation and saved checkpoints.
- $\pi_{0.5}$ (Physical Intelligence et al., 2025). The policy conditions on the current scene and wrist images together with the task instruction and current torque vector. Its image frontend resizes the inputs to 224×224 . The wrapper discretizes the normalized torque values into 256 bins and includes

them in the language prompt. The action expert predicts a 36-action chunk by flow matching, using ten Euler steps at inference. The expert-only configuration freezes the vision and language backbone while training the action expert and action/time projections. Its task configurations use AdamW with learning rate 2.5×10^{-5} and weight decay 10^{-4} , and gradient accumulation to reach effective batch size 256.

D.4 CLOSED-LOOP EVALUATION

For Section 4.4, the robot sends its observation window to a remote inference machine, receives an action chunk, and executes it at 30 Hz under the receding-horizon schedule. A human evaluator labels each episode as success or failure, using the same criteria used during teleoperation. Each method is evaluated in 30 trials per task.

D.5 REPRESENTATION DIAGNOSTICS AND EVALUATION

For visualization, we fit a pixel decoder d_ψ to frozen encoder outputs $\mathbf{z}_t = \text{Enc}(\mathbf{o}_t)$. Learnable per-patch queries use cross-attention to map these representations to pixel patches. Only the decoder parameters ψ are trained, using per-pixel MSE over training frames:

$$\mathcal{L}_{\text{dec}}(\psi) = \mathbb{E}_{\mathbf{o}_t} [\text{MSE}(d_\psi(\text{Enc}(\mathbf{o}_t)), \mathbf{o}_t)] . \quad (12)$$

The encoder and latent dynamics remain frozen. Framewise reconstructions are $d_\psi(\mathbf{z}_t)$, and decoded rollout predictions are $d_\psi(\hat{\mathbf{z}}_t)$.

Figure 16 compares both encoder variants on the same held-out Pick and Place T episode from the scene camera. Each strip shows recorded video, framewise reconstructions, and decoded autoregressive predictions. Latent rollouts start from the initial observation context and advance under recorded actions, without intermediate ground-truth frames or newly generated actions. We display ten frames at indices 0, 3, 6, 9, 12, 15, 18, 21, 24, and 26, spanning 10.4 seconds at the 2.5 Hz model sampling rate.

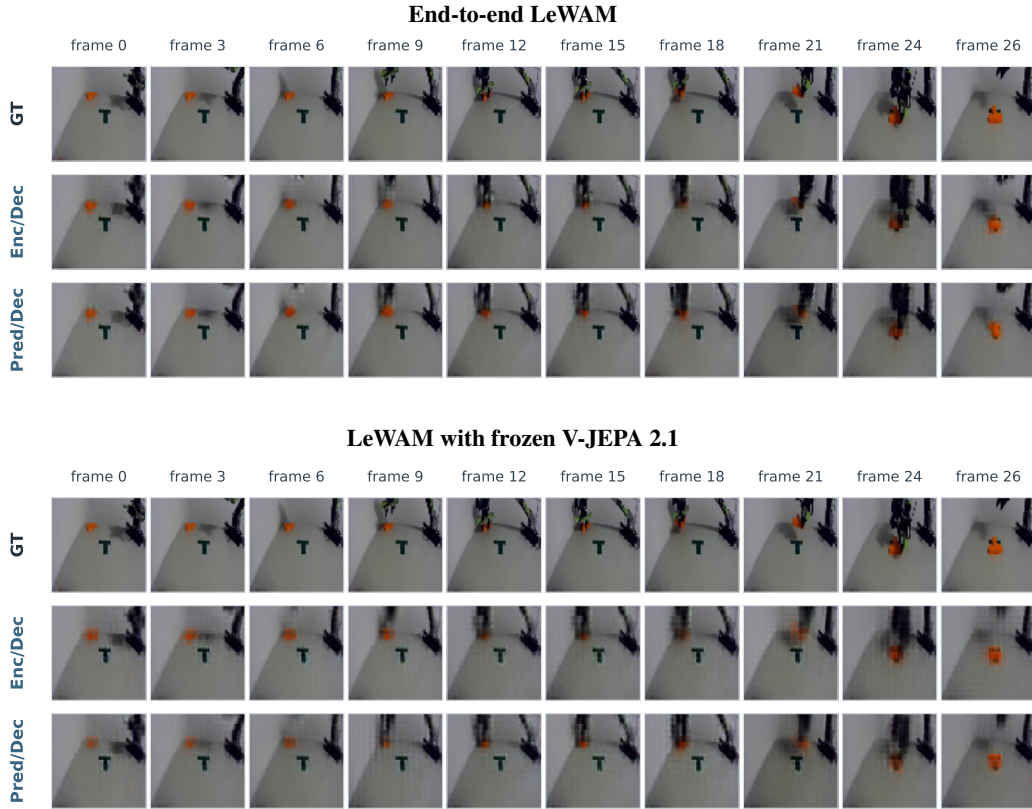


Figure 16: Matched visual reconstructions and recorded-action latent rollouts for the scene camera. Within each model, rows show the ground-truth (GT) video, independent frame reconstructions, and decoded autoregressive predictions.

E DISCUSSION OF THE ROLE OF JOINT PREDICTION

The conditional-risk analysis describes how action supervision shapes representations; the planning discussion connects learned proposals to contact-dependent outcomes. These mechanisms complement Sections 4 and 3.2.

Table 13 complements Section 2 with a comparison of training requirements and prediction capabilities.

Table 13: **Modeling and Control Capabilities.** FDM denotes forward dynamics conditioned on clean actions; Policy denotes learned action prediction. Entries describe reported training recipes or the goal-reaching configurations used here.

Method	FDM	Policy	Joint FDM and Policy	End-to-end from scratch
<i>World models</i>				
DINO-WM (Zhou et al., 2024)	✓	✗	✗	✗
V-JEPA 2-AC (Assran et al., 2025)	✓	✗	✗	✗
LeWM (CEM) (Maes et al., 2026b)	✓	✗	✗	✓
<i>World action models</i>				
UVA (Li et al., 2025)	✓	✓	✗	✗
UWM (Zhu et al., 2025)	✓	✓	✗	✗
DreamZero (Ye et al., 2026)	✗	✓	✗	✗
Fast-WAM (Yuan et al., 2026)	✗	✓	✗	✗
Cosmos 3 (NVIDIA, 2026)	✓	✓	✗	✗
LeWAM (ours)	✓	✓	✓	✓
<i>Goal-conditioned policy</i>				
LeWM (IDM) (Nguyen et al., 2026)	✓	✓	✗	✗
GCBC (Ghosh et al., 2021)	✗	✓	✗	✗

Joint requires clean-action FDM and policy losses on the same sample in one training pass. From scratch includes visual encoding and prediction without pretrained or frozen components. UWM uses a frozen pretrained variational autoencoder even when its robot backbone is trained from scratch.

E.1 REPRESENTATION AND DYNAMICS

Representation Shaping. SIGReg (Balestriero & LeCun, 2025; Maes et al., 2026b) regularizes the latent marginal; action supervision encourages control-relevant distinctions (Figure 17). If visual history changes the conditional action velocity, discarding it increases optimal action regression risk. Latent prediction adds pressure to retain action-conditioned consequences.

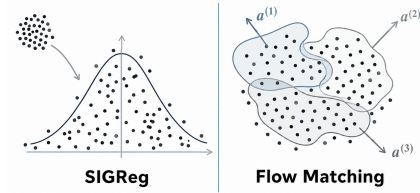


Figure 17: Conceptual roles of SIGReg and action flow matching in shaping the latent space.

Conditional Action Risk. In Equation 2, squared flow matching makes the optimal velocity the conditional mean of $\mathbf{a}_j - \epsilon_j$. Let $\mathcal{I}_j$ include the noisy action prefix, flow time, and other conditioning, but exclude visual history $\mathbf{z}_{\leq t}$. With $\mathcal{R}_{\text{FM}}$ denoting optimal velocity MSE, the conditional projection identity gives

$$\begin{aligned} & \mathcal{R}_{\text{FM}}(\text{constant}) - \mathcal{R}_{\text{FM}}(\mathbf{z}_{\leq t}) \\ &= \frac{1}{C} \sum_{j=t}^{t+C-1} \mathbb{E}[\text{MSE}(\mathbb{E}[\mathbf{a}_j - \epsilon_j \mid \mathcal{I}_j, \mathbf{z}_{\leq t}], \mathbb{E}[\mathbf{a}_j - \epsilon_j \mid \mathcal{I}_j])] \geq 0. \end{aligned} \quad (13)$$

When visual history changes the conditional velocity, a constant code incurs higher optimal risk. Action supervision therefore favors these distinctions, while SIGReg regularizes the global latent distribution.

E.2 PLANNING WITH LEARNED ACTION PROPOSALS

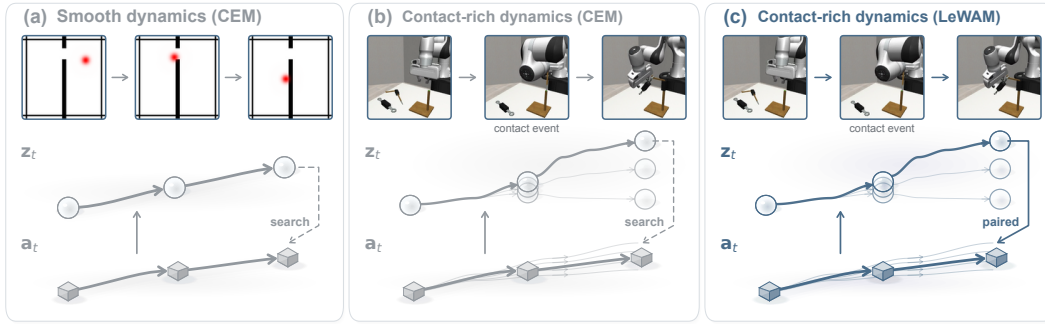


Figure 18: **Action Proposals and Predicted Consequences.** (a) Smooth rollouts; (b) candidate rollouts branch at contact; (c) LeWAM pairs learned action proposals with predicted outcomes. Dashed links denote search; the solid link denotes joint prediction.

Ill-Posed Action Solver in Contact-rich Events. Prior JEPAs for control *e.g.*, DINO-WM (Zhou et al., 2024) and LeWM (Maes et al., 2026b) learn action-conditioned latent dynamics without an action-prediction objective and uses CEM (Rubinstein, 1999) for downstream control. In smooth dynamics like 2D navigation, the action solver is *well-posed*: a diverse representation can invert the action trajectories by argmin the cost of action-conditioned dynamics. However, a contact event can switch the active mode, so nearby actions yield distinct outcomes (Pang et al., 2023). If sampled plans miss useful contact, an object-based goal cost provides little ranking signal (Yamada et al., 2024). With a finite population, CEM ranks the contact branches it samples, and additional iterations incur more rollouts.

Behavior and Consequence. Joint prediction additionally supplies a learned action distribution for planning proposals. Learned proposals can improve contact-rich planning under constrained budgets (Brudermüller et al., 2025). Figure 18 illustrates why the population-based search from a broad prior cannot work on contact-rich manipulation, and how LeWAM uses the learned proposals and predicted consequences to alleviate this problem. The planning objective is defined in Section 3.2, and the refinement procedure is specified in Appendix A.1. Though this problem in 3D contact-rich manipulation is well known and our solution is heuristic, we consider the rigorous verification of the claim as future work.

F AUTHOR CONTRIBUTION STATEMENT

All coauthors contributed to project discussions, writing, and editing.

Minghao initiated the project and led its technical development. He proposed the general architecture with action flow matching, MoT, the initial version of gradient-based planning, and constructed the benchmarks. He also wrote the main part of the paper and built the webpage.

Tavis designed the details of model architecture and MoT attention mechanism, optimized the planning algorithm to use batched gradients, and refined the methodology to the final version through extensive experimentation. He also finalized the released code. Tavis and Minghao completed the simulation experiments reported in the paper.

Eryk designed and carried out the entire physical robot study. He built the hardware and teleoperation pipeline, adapted the model for real-robot deployment, collected all real-robot data, trained models for real robots, and ran the physical evaluations. He also wrote the main part of physical-robot experiments section and prepared its results.

Randall initiated and supervised the project and guided its overall direction. He advised on representation learning, experimental design, and result verification, contributed extensively to writing and refining the manuscript, and finalized the paper’s title, narrative, and branding.